

MITALI PATLE

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Education

Amrita Viswa Vidyapeetham

August 2023 – Present

Bachelor of Technology in Computer Science (Current CGPA: 7.18)

Coimbatore, TN

Areas of Interest

- Embedded Systems
- Internet of Things
- Backend development

Experience

KodeAstra | *Python, FastAPI, LangChain, LangGraph, LLMs*

April 2026 – May 2026

- Architected an AI-powered NCLEX question paper generation workflow for B2B deployment, enabling coaching academies to programmatically configure and export exam-ready papers
- Designed REST APIs and frontend state management for draft editing, change preview, and permission-scoped access control
- Integrated a GPT-powered AI editing assistant into a Plate.js-based rich text editor, returning validated structured edits from natural-language prompts with apply/undo workflow

Projects

SkillExchange | *React.js, Node.js, Express.js, MongoDB*

- Built a full-stack peer skill-exchange platform with React — responsive UI, Axios async fetching, and state-driven updates for dynamic skill request, offer, and match listings.
- Designed RESTful APIs with Express and Mongoose for user, skill, and request management — secured via JWT authentication, bcrypt hashing, and protected API routes with validation and error handling.
- Structured MongoDB schema with targeted indexing to optimize query efficiency and reduce API response latency.

Expense Tracker App | *Flutter, Dart, fl chart*

- Built a mobile expense tracking application to record and categorize daily financial transactions.
- Implemented data aggregation logic in Dart to compute total spend, category-wise expenses, and daily averages in real time.
- Integrated fl chart to visualize monthly spending trends using dynamic bar and pie charts.
- Designed modular analytics screen for grouped expense breakdown and responsive dashboard updates.

LoRa-Based Wearable Health Monitoring | *STM32 · MAX30102 · SX1278 LoRa · Edge ML*

- Built a 3-layer LoRa (433 MHz) IoT network (Tag → Relay → Gateway) using STM32, SX1278, and MAX30102 for real-time BPM/SpO₂ monitoring in connectivity-denied environments.
- Implemented Edge ML anomaly detection on STM32 using EWMA, Z-score analysis, and sliding-window voting for real-time ALERT, SOS, and MISSING event detection.
- Developed real-time STM32 firmware with I2C/SPI communication, LoRa routing, RSSI-based relay selection, failover, and non-blocking scheduling for low-latency operation.

Technical Skills

Programming Languages C++, Java, JavaScript/TypeScript, Python, Dart, SQL

Backend Development Node.js, Express.js, REST API Design

Frontend & Mobile React.js, React Native, Flutter

Microcontrollers STM32, ESP32, Arduino

Databases MongoDB, SQLite, PostgreSQL

Tools Git, Postman, VS Code

Relevant Coursework

- Internet of Things
- Database Management Systems
- Operating Systems
- Computer Networks
- Embedded Systems
- Object Oriented Programming

Extracurriculars

Event Coordinator: Led campus decoration. Department secured 1st place.

Newsletter Editor, Anokha Tech Fair 2024: Drafted and edited project showcase newsletter.

National Service Scheme Volunteer (2 Years): Assisted in outreach activities and school engagement programs.

Class Representative (2 Consecutive Years): Acted as a liaison between faculty and students, coordinated academic communications, and supported class-level organization.